

Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning

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Binary Mixture chosen: Benzene – Toluene

Thermodynamic Model Chosen: Peng–Robinson (PR)

Machine Learning Methods Compared: Polynomial Regression, Random Forest, Support Vector Regression (SVR)

1. Problem Statement

Binary distillation is a widely used separation process in chemical engineering. However, repeated rigorous simulations can become computationally expensive when a large number of operating conditions need to be evaluated. The objective of this project is to develop a machine-learning-based surrogate model that approximates the behaviour of a binary distillation column using data generated from DWSIM simulations.

A benzene–toluene binary mixture was selected for the study. The distillation column was simulated in DWSIM using the **Peng–Robinson (PR)** thermodynamic model. A dataset containing **100 successful simulation cases** was generated by varying selected operating conditions, primarily feed temperature, feed pressure, reflux ratio, and bottoms withdrawal rate.

The surrogate model uses the selected operating conditions as inputs and predicts four important column performance variables:

- Distillate purity (x_D)
- Bottoms purity (x_B)
- Condenser heat duty (QC)
- Reboiler heat duty (QR)

Three machine learning approaches were implemented and compared:

- Polynomial Regression
- Random Forest
- Support Vector Regression (SVR)

The models were evaluated using **R², mean absolute error (MAE), and root mean square error (RMSE)**. In addition to the test-set evaluation, **5-fold cross-validation** and physical consistency checks were performed to assess model stability and whether the predictions remained within physically meaningful ranges.

2. DWSIM Flowsheet and Model Description

The process was modelled as a binary distillation system in DWSIM using a benzene–toluene mixture. The **Peng–Robinson equation of state** was used as the thermodynamic property model.

The flowsheet consists of a feed stream entering the distillation column and separate distillate and bottoms product streams. The column includes a condenser and reboiler, allowing both separation performance and energy requirements to be obtained from each simulation.

The basic column configuration was kept consistent between simulation cases while selected operating conditions were changed. For each successful simulation, the operating conditions and resulting column outputs were recorded automatically into a CSV dataset.

The final dataset contains **100 simulation cases and 11 columns**. Seven columns represent the selected input variables and four columns represent the target outputs.

The DWSIM simulations serve as the reference data for training and evaluating the surrogate models. The purpose of the machine learning models is to reproduce these simulation results without requiring a new rigorous DWSIM calculation for every prediction.

3. Input and Output Variables

The surrogate modelling problem was formulated by using selected operating conditions as input features and four column performance variables as prediction targets.

3.1 Input Variables

The seven input variables recorded in the dataset are:

Input Variable	Description	Unit
Feed Temperature	Temperature of the feed entering the column	K
Feed Pressure	Pressure of the feed stream	Pa
Feed Benzene Mole Fraction	Benzene mole fraction in the binary feed	dimensionless
Number of Stages	Number of theoretical stages in the column	stages
Feed Stage	Stage at which the feed enters the column	stage
Reflux Ratio	Ratio of liquid reflux returned to the column relative to distillate withdrawal	dimensionless
Bottoms Withdrawal Rate	Molar flow rate withdrawn through the bottoms stream	mol/s

In the final dataset, the **feed benzene mole fraction was fixed at 0.50**, the **number of stages was fixed at 10**, and the **feed stage was fixed at 5**. Variation was introduced through feed temperature, feed pressure, reflux ratio, and bottoms withdrawal rate.

This means that the dataset represents a restricted operating region rather than the full design space of the distillation column. The three fixed variables are still retained as input columns because they are part of the specified operating conditions and the dataset structure.

3.2 Output Variables

Four outputs were selected as the targets of the surrogate model:

Output Variable	Description
Distillate Purity (xD)	Purity/composition measure obtained from the distillate stream
Bottoms Purity (xB)	Purity/composition measure obtained from the bottoms stream
Condenser Heat Duty (QC)	Heat removed by the condenser
Reboiler Heat Duty (QR)	Heat supplied to the reboiler

The final dataset therefore contains **7 input variables and 4 output variables**, giving a total of **11 columns and 100 simulation cases**.

4. Dataset Generation Strategy

The dataset was generated directly from successful DWSIM simulations. A total of **100 cases** were used for model development.

For each case, the selected operating conditions were applied to the feed and column, the DWSIM calculation was solved, and the resulting purity and heat-duty values were recorded automatically.

The variables varied during dataset generation were:

- Feed temperature
- Feed pressure
- Reflux ratio
- Bottoms withdrawal rate

The feed composition, number of stages, and feed stage were kept fixed. This provided a controlled operating region while avoiding unnecessary changes to the basic column configuration.

The final CSV dataset was inspected using Python and Pandas. The inspection confirmed:

- **Dataset size:** 100 rows $\times$ 11 columns

- **Missing values:** 0
- **Duplicate rows:** 0
- **Data types:** numerical for all variables

All 100 generated cases were therefore retained for machine learning.

5. Range of Operating Conditions

The final dataset covers the following operating ranges:

Variable	Minimum	Maximum
Feed Temperature	351.15 K	355.90 K
Feed Pressure	99,000 Pa	102,000 Pa
Feed Benzene Mole Fraction	0.50	0.50
Number of Stages	10	10
Feed Stage	5	5
Reflux Ratio	3.00	4.35
Bottoms Withdrawal Rate	13.70 mol/s	13.97 mol/s

The corresponding DWSIM output ranges were:

Output	Minimum	Maximum
Distillate Purity (xD)	0.9352	0.9525
Bottoms Purity (xB)	0.9408	0.9618
Condenser Duty (QC)	1725.65	2317.82
Reboiler Duty (QR)	1791.70	2390.52

These ranges define the operating region over which the surrogate model was developed and evaluated. Predictions substantially outside these ranges should be treated cautiously because they would involve extrapolation beyond the DWSIM data used for model development.

6. Data Preprocessing

The generated CSV dataset was inspected using Python and Pandas before model training.

The inspection confirmed:

- **100 observations**
- **11 variables**
- **0 missing values**
- **0 duplicate rows**
- Numerical data types for all variables

The seven operating variables were separated from the four target variables. The four targets were treated as separate regression problems:

- xD
- xB
- QC
- QR

The dataset was divided into an **80/20 training-test split**, giving:

- **80 training cases**
- **20 testing cases**

The training data was used to fit the models, while the test data was kept separate for evaluating predictions on unseen cases.

For Polynomial Regression, polynomial features were generated from the input variables before fitting the regression model. Random Forest was trained using the numerical input features directly. For SVR, feature scaling was applied because the input variables have substantially different numerical scales, particularly pressure and temperature.

7. Machine Learning Methods

Three machine learning methods were implemented and compared: **Polynomial Regression, Random Forest, and Support Vector Regression (SVR)**.

7.1 Polynomial Regression

Polynomial Regression was used to capture nonlinear relationships between the operating conditions and the column outputs.

This approach was appropriate for the dataset because the relationship between operating conditions and distillation performance is not necessarily linear.

Polynomial features allow the regression model to represent nonlinear trends and interactions between the input variables.

A separate Polynomial Regression model was trained for each of the four target variables.

7.2 Random Forest

Random Forest is an ensemble method based on multiple decision trees. It can capture nonlinear relationships and interactions between operating variables without requiring an explicit mathematical relationship between the inputs and outputs.

The Random Forest model was trained using the same input-output structure as the other models so that its performance could be compared directly.

7.3 Support Vector Regression

Support Vector Regression was selected as a third modelling approach based on a different regression strategy.

SVR attempts to construct a function that predicts the target while controlling prediction error and model complexity. Since the input variables have different numerical scales, feature scaling was applied before training the SVR models.

Separate SVR models were trained for xD, xB, QC, and QR.

8. Model Training and Evaluation

The models were evaluated using three standard regression metrics.

R²

R^2 measures how well the model explains the variation in the target variable. A value close to 1 indicates that the predictions closely follow the reference values.

MAE

Mean Absolute Error (MAE) measures the average absolute difference between the predicted value and the corresponding DWSIM value.

RMSE

Root Mean Square Error (RMSE) gives greater weight to larger prediction errors and therefore helps identify models that occasionally produce larger deviations.

An **80/20 train-test split** was used for the primary evaluation. In addition, **5-fold cross-validation** was performed to assess whether model performance remained consistent across different subsets of the available data.

The evaluation was performed separately for all four target variables.

9. Performance Comparison

The test-set results were:

Output	Model	R^2	MAE	RMSE
Distillate Purity	Polynomial Regression	0.999963	0.000032	0.000040
Distillate Purity	Random Forest	0.966626	0.000716	0.001207
Distillate Purity	SVR	0.884340	0.001705	0.002248
Bottoms Purity	Polynomial Regression	0.999968	0.000025	0.000033
Bottoms Purity	Random Forest	0.998354	0.000196	0.000236
Bottoms Purity	SVR	0.967375	0.000820	0.001050
Condenser Duty	Polynomial Regression	1.000000	0.002960	0.003770
Condenser Duty	Random Forest	0.999117	4.925820	6.642152
Condenser Duty	SVR	0.929088	40.236028	59.515579

Output	Model	R ²	MAE	RMSE
Reboiler Duty	Polynomial Regression	1.000000	0.002776	0.003669
Reboiler Duty	Random Forest	0.998368	6.848227	8.983348
Reboiler Duty	SVR	0.925526	40.934649	60.686081

Polynomial Regression produced the best test-set performance for **all four target variables**.

For distillate purity, Polynomial Regression achieved an R² of **0.999963**, compared with 0.966626 for Random Forest and 0.884340 for SVR.

For bottoms purity, Polynomial Regression achieved an R² of **0.999968**.

For condenser and reboiler duties, Polynomial Regression achieved R² values effectively equal to **1.000000**, with RMSE values of only **0.003770** and **0.003669**, respectively.

Random Forest produced the second-best overall performance. Its predictions were particularly strong for bottoms purity and the two heat-duty targets, although its errors remained higher than those of Polynomial Regression.

SVR produced the weakest overall results. Its performance was particularly lower for the two heat-duty targets, where its RMSE values were approximately 59.52 and 60.69.

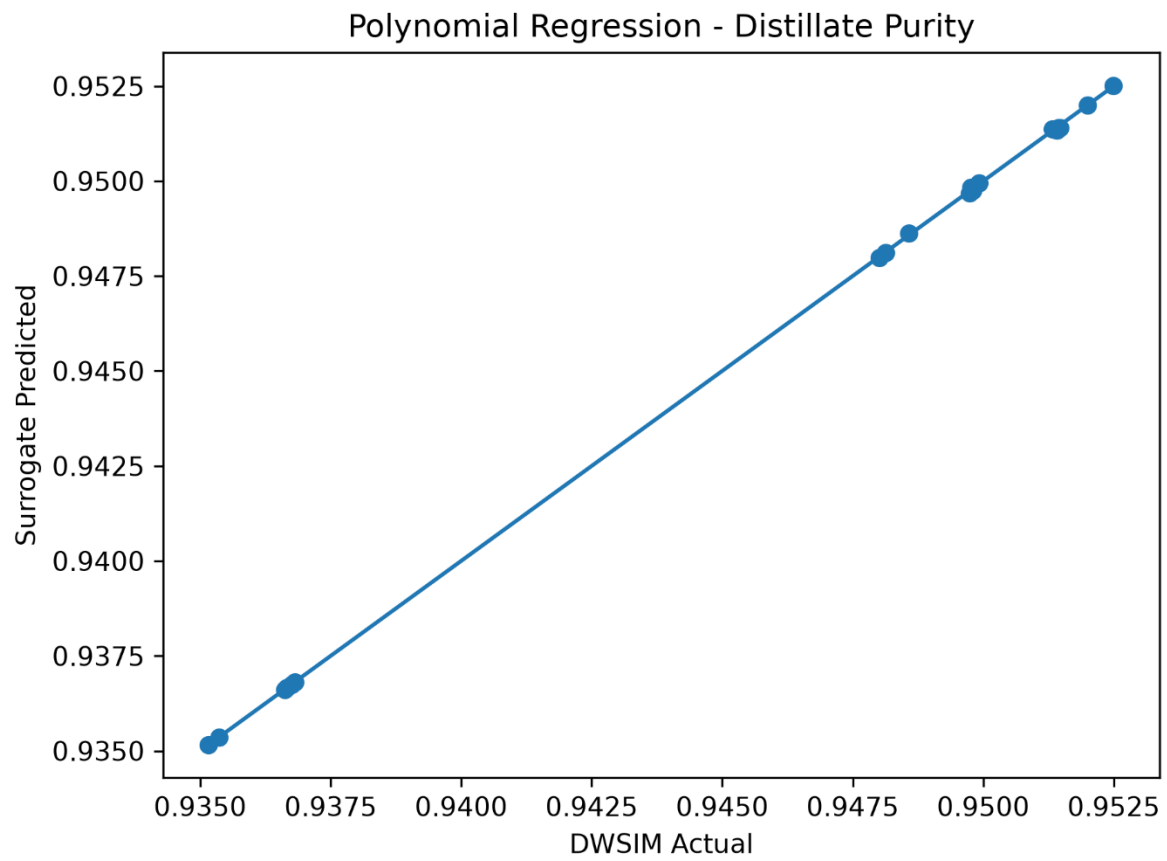


Figure 1. Predicted versus actual distillate purity (xD) for Polynomial Regression.

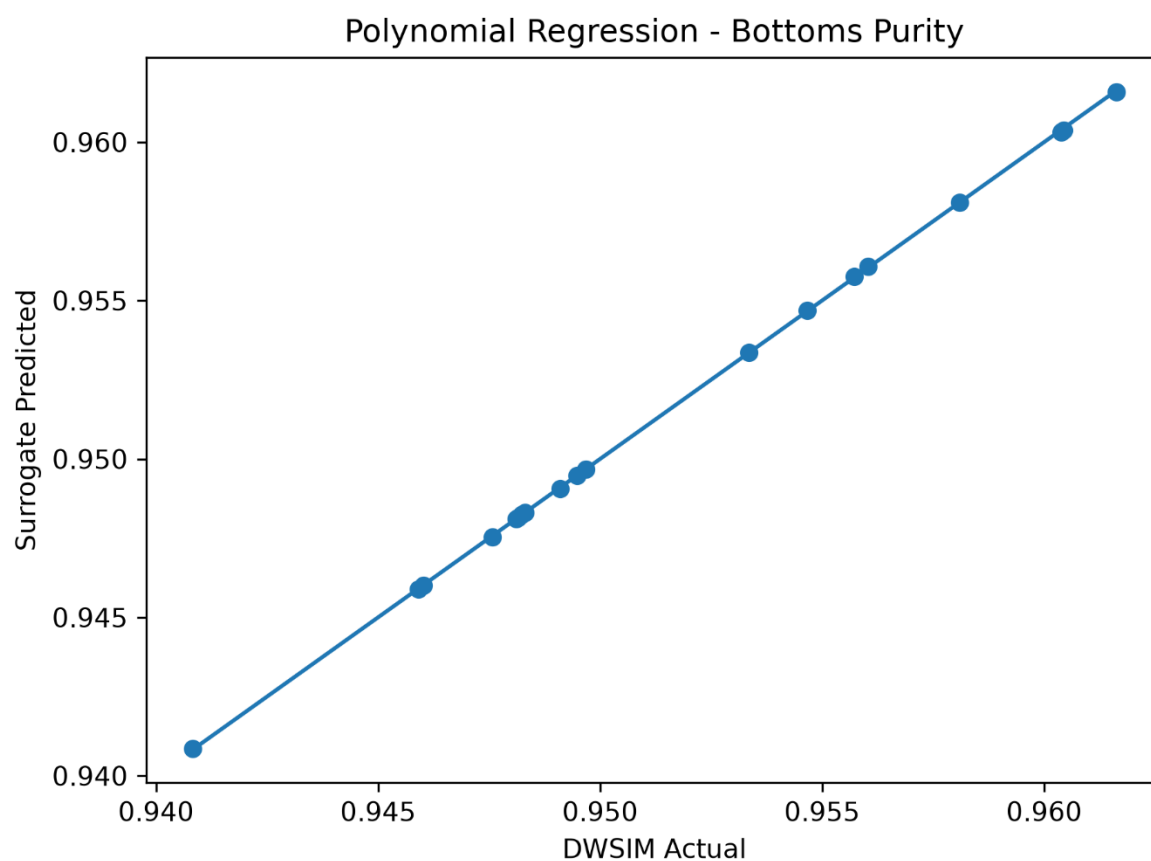


Figure 2. Predicted versus actual bottoms purity (xB) for Polynomial Regression.

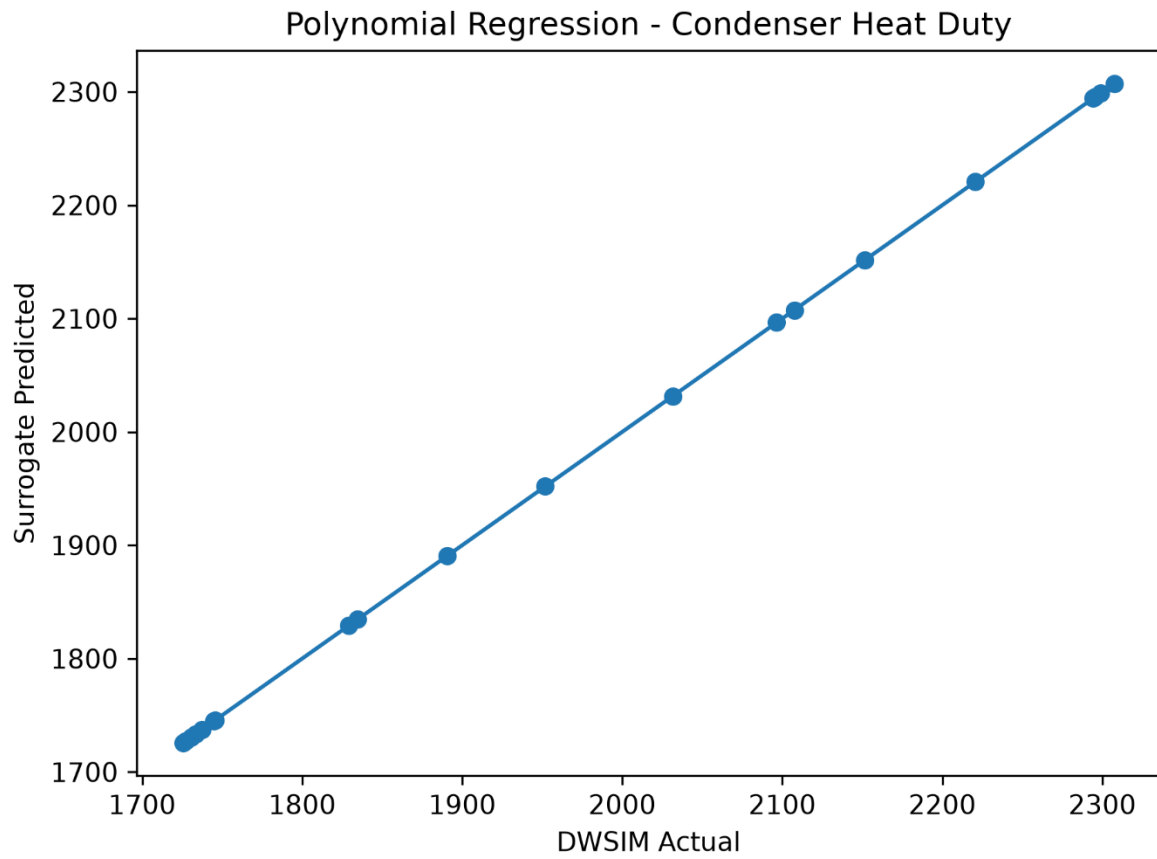


Figure 3. Predicted versus actual condenser heat duty (QC) for Polynomial Regression.

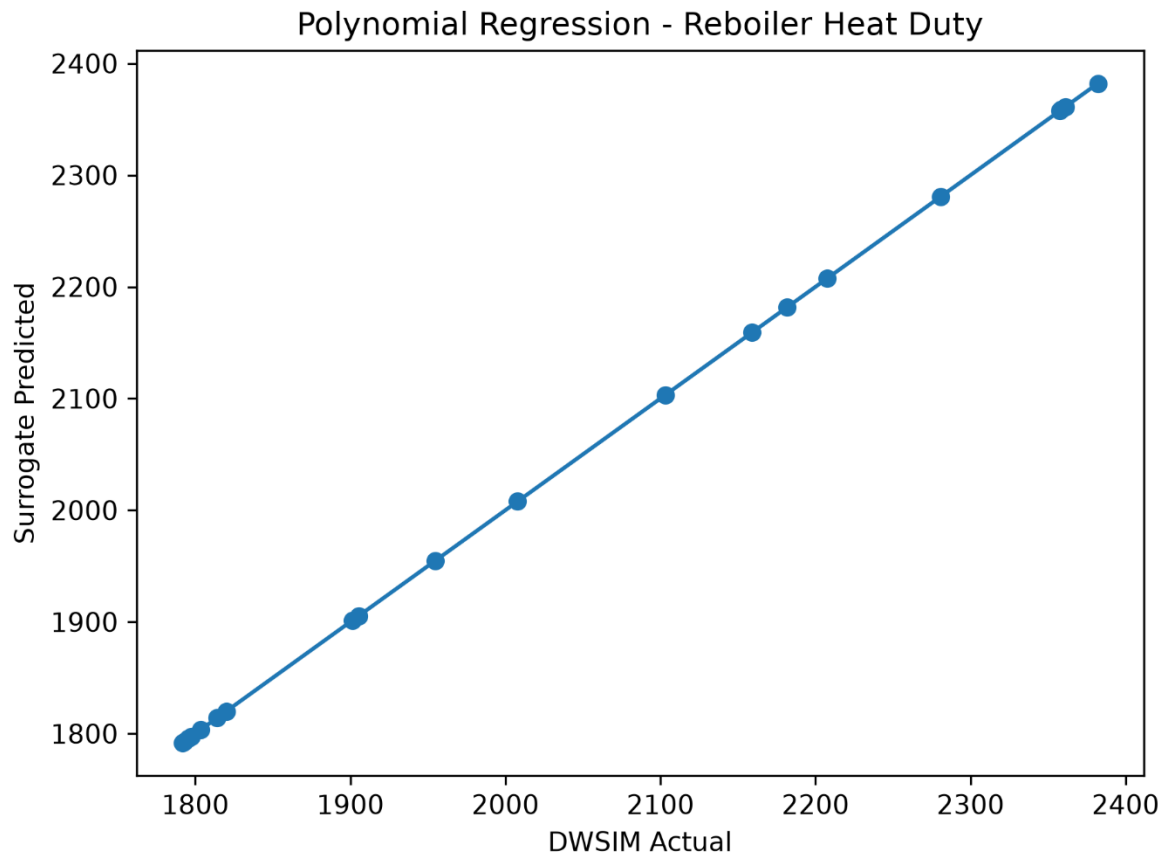


Figure 4. Predicted versus actual reboiler heat duty (QR) for Polynomial Regression.

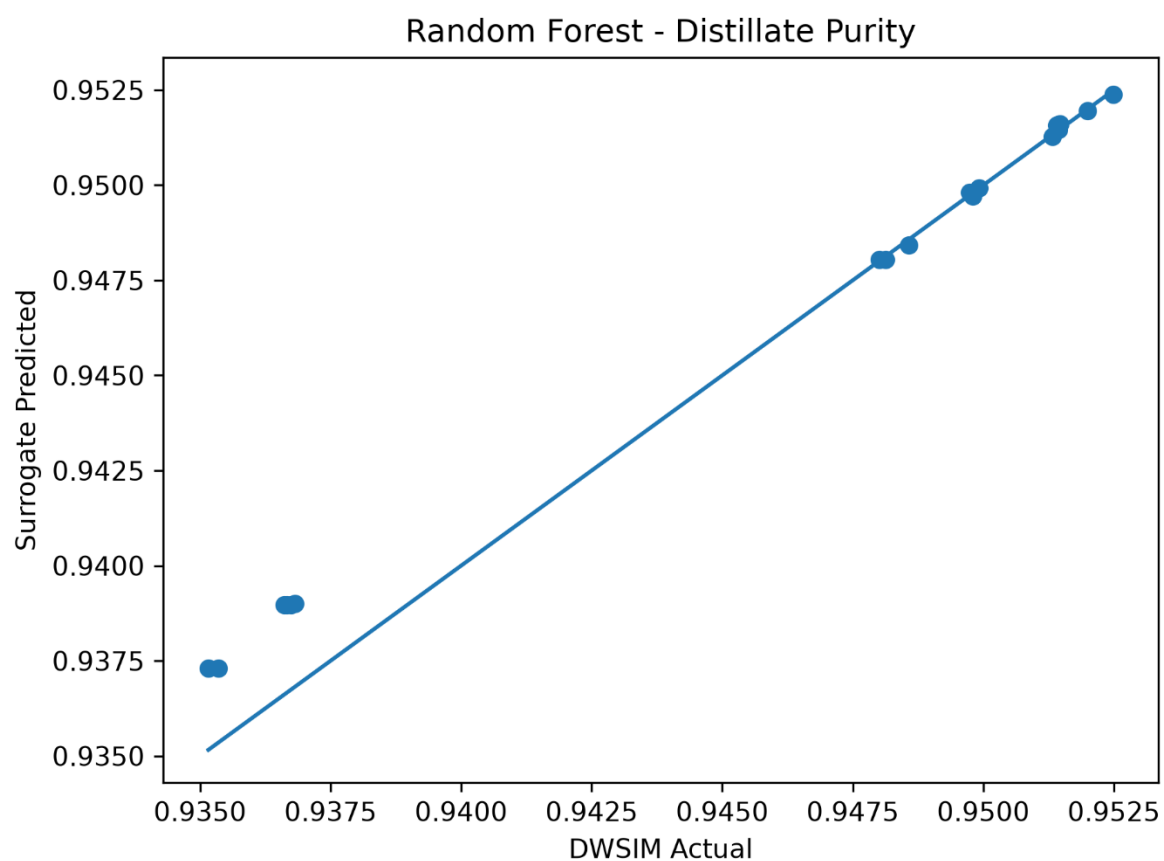


Figure 5. Predicted versus actual distillate purity (x_D) for Random Forest.

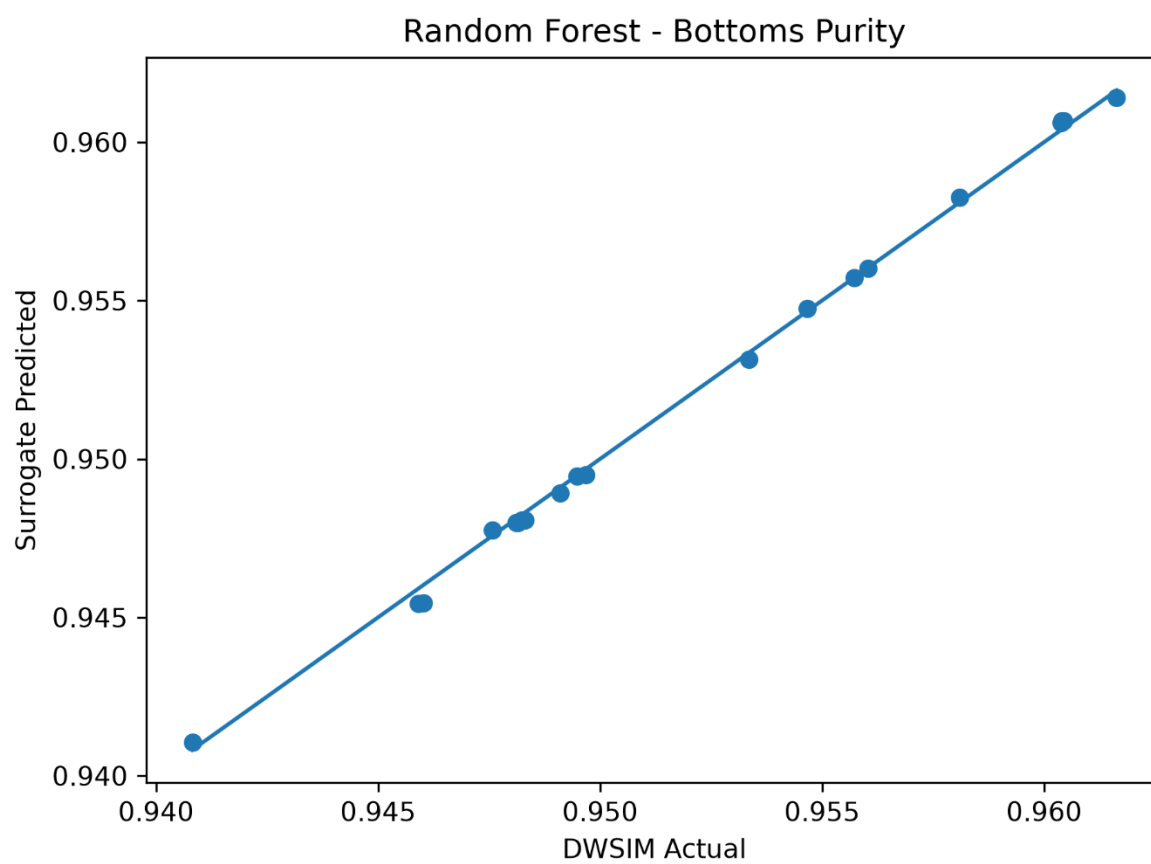


Figure 6. Predicted versus actual bottoms purity (xB) for Random Forest.

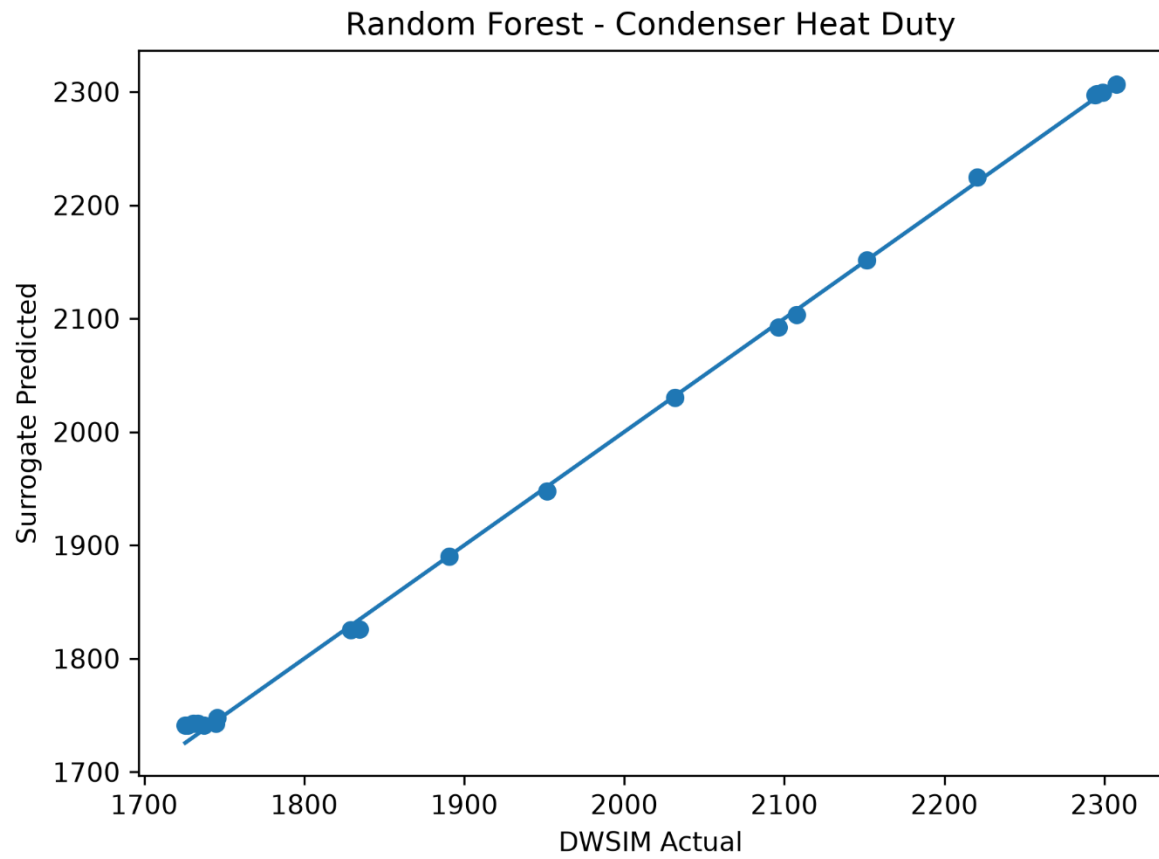


Figure 7. Predicted versus actual condenser heat duty (QC) for Random Forest.

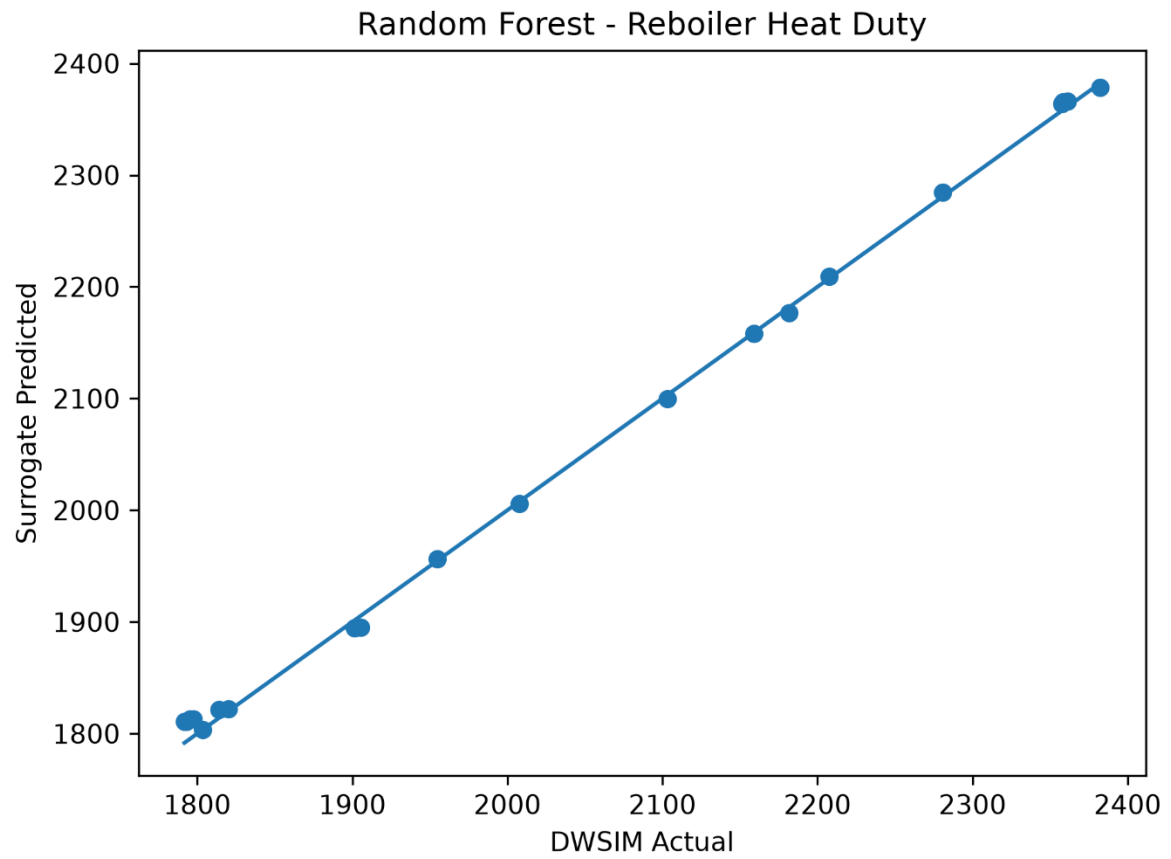


Figure 8. Predicted versus actual reboiler heat duty (QR) for Random Forest.

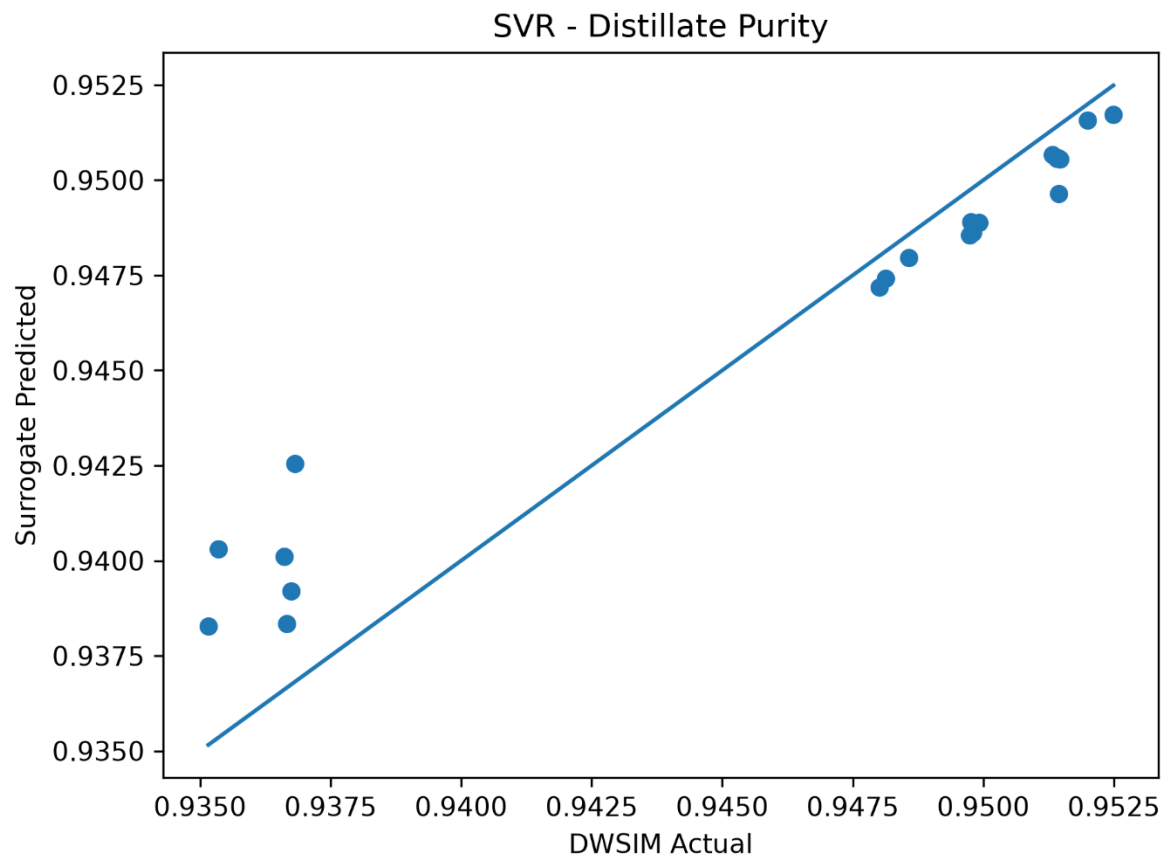


Figure 9. Predicted versus actual distillate purity (xD) for Support Vector Regression (SVR).

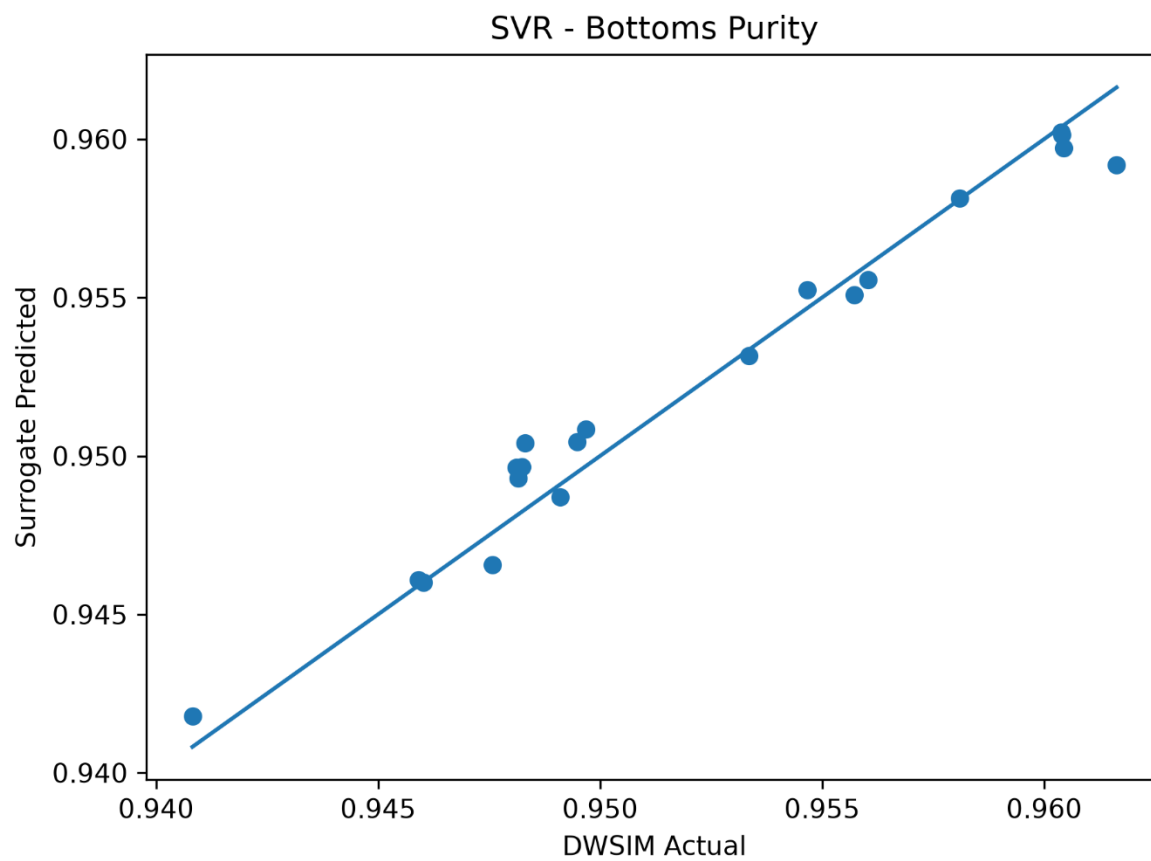


Figure 10. Predicted versus actual bottoms purity (xB) for Support Vector Regression (SVR).

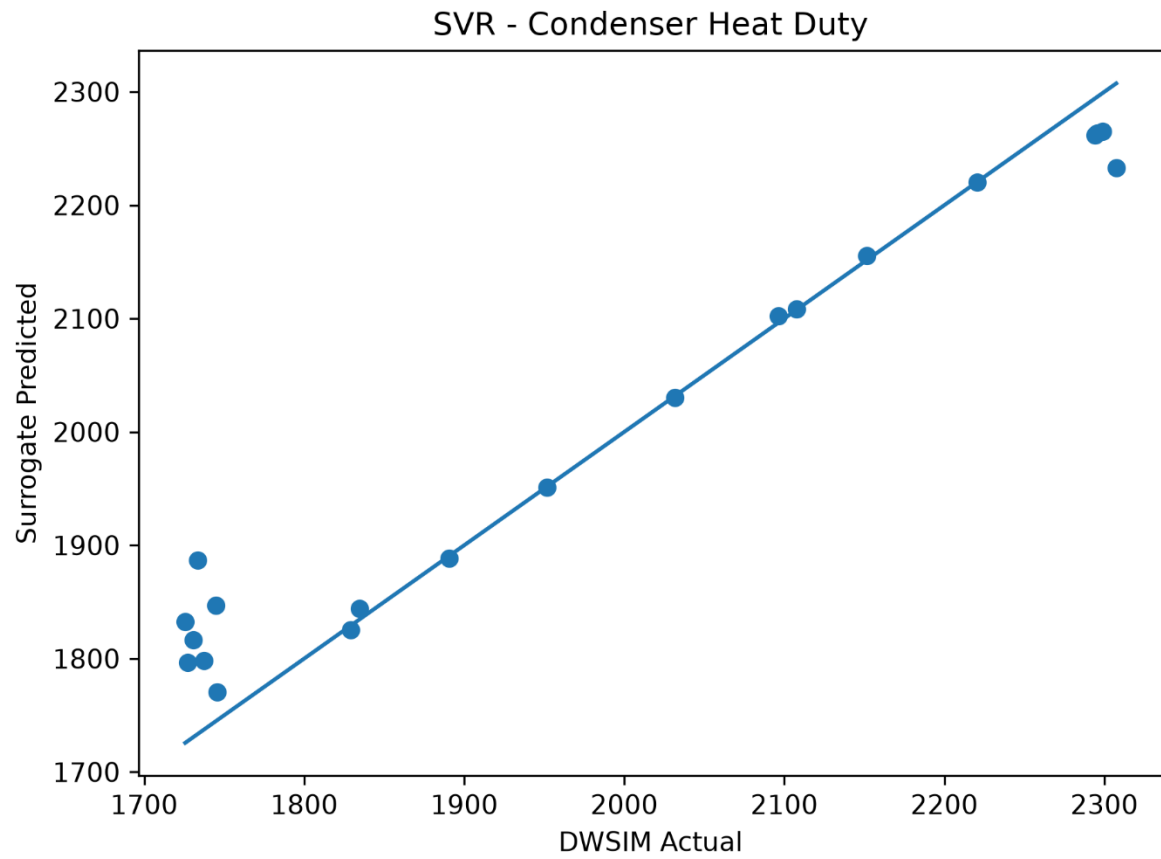


Figure 11. Predicted versus actual condenser heat duty (QC) for Support Vector Regression (SVR).

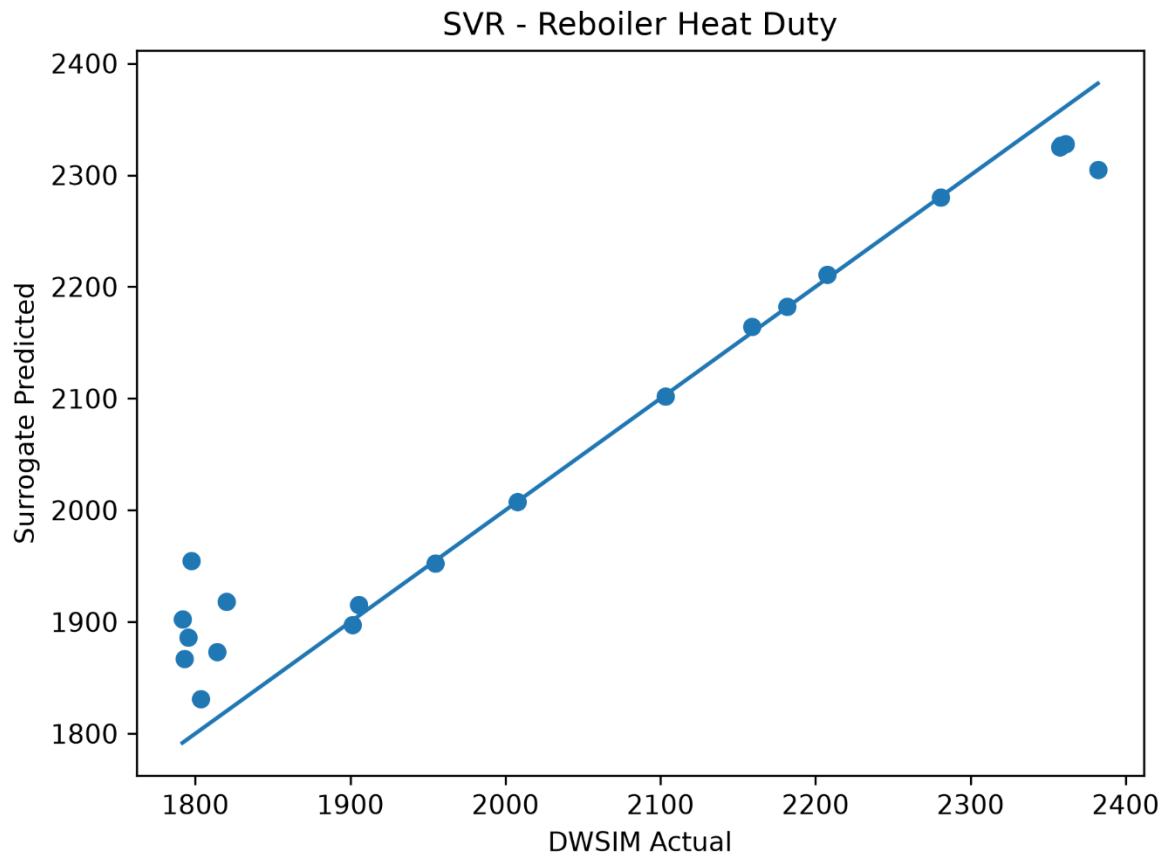


Figure 12. Predicted versus actual reboiler heat duty (QR) for Support Vector Regression (SVR).

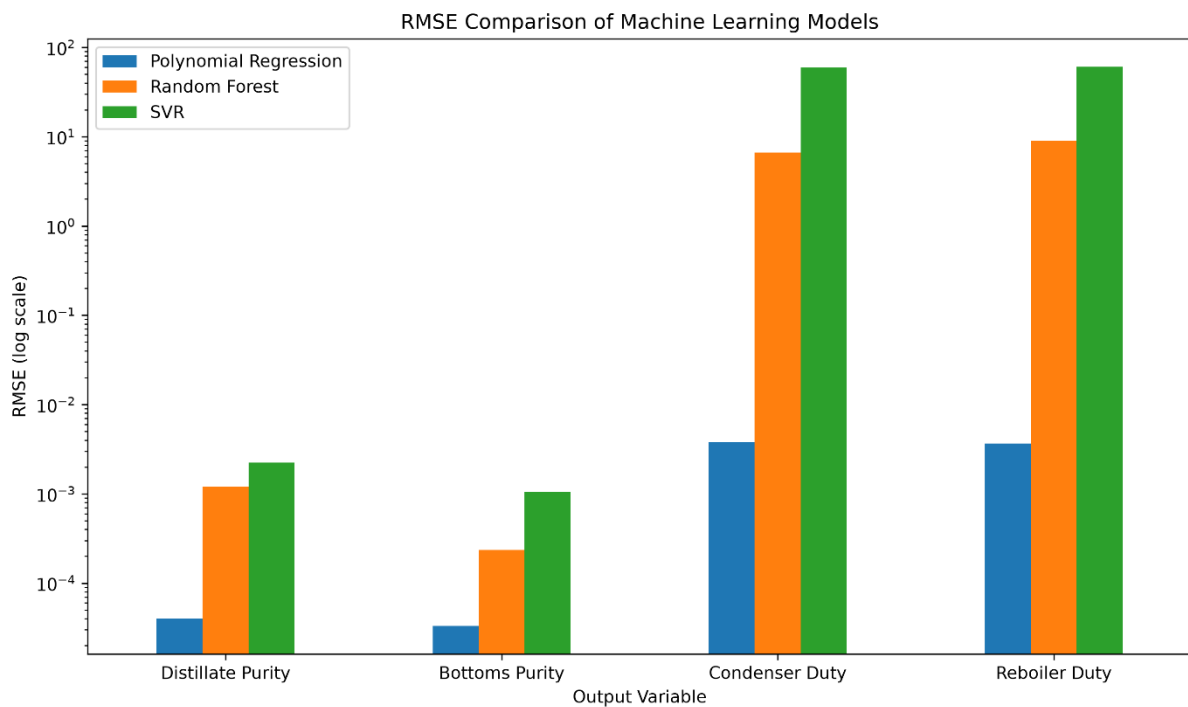


Figure 13. RMSE comparison of Polynomial Regression, Random Forest, and Support Vector Regression across the four target variables.

10. Cross-Validation and Robustness

To determine whether the high test-set performance depended on a particular train-test split, **5-fold cross-validation** was performed.

The mean R^2 values were:

Output	Polynomial Regression	Random Forest	SVR
Distillate Purity	0.999716	0.995096	0.878893
Bottoms Purity	0.999966	0.996331	0.962227
Condenser Duty	1.000000	0.998886	0.967121
Reboiler Duty	1.000000	0.997955	0.966000

Polynomial Regression maintained a mean R^2 above **0.9997 for every target**.

Its cross-validation standard deviations were:

Output	Polynomial Regression R^2 Standard Deviation
Distillate Purity	0.000391
Bottoms Purity	0.0000067
Condenser Duty	1.98×10^{-10}
Reboiler Duty	1.96×10^{-10}

These small variations indicate that Polynomial Regression produced highly consistent results across the five folds.

Random Forest also demonstrated good robustness, with mean R^2 values above 0.995 for all four outputs.

SVR showed considerably greater variation for distillate purity. Its mean R^2 was **0.878893**, with a standard deviation of approximately **0.102**. This indicates that its performance was more sensitive to the particular subset of training data.

Overall, the cross-validation results support the test-set comparison and provide evidence that the strong Polynomial Regression performance was consistent across different data splits.

11. Physical Consistency of Predictions

Statistical accuracy alone is not sufficient for evaluating a chemical engineering surrogate model. The final predictions were therefore checked for basic physical consistency.

The physical validation produced the following results:

Check	Valid Predictions
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Distillate Purity	100 / 100
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Bottoms Purity	100 / 100
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Condenser Duty	100 / 100
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Reboiler Duty	100 / 100
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The predicted ranges were also very close to the corresponding DWSIM ranges.

Distillate Purity

DWSIM range:

0.9351597 – 0.9525357

Surrogate range:

0.9351609 – 0.9525676

Bottoms Purity

DWSIM range:

0.9407915 – 0.9617605

Surrogate range:

0.9408035 – 0.9617350

Condenser Duty

DWSIM range:

1725.6547 – 2317.8165

Surrogate range:

1725.6602 – 2317.8064

Reboiler Duty

DWSIM range:

1791.6962 – 2390.5186

Surrogate range:

1791.7022 – 2390.5077

The predicted ranges are therefore extremely close to the corresponding DWSIM ranges, indicating that the surrogate did not produce predictions far outside the behaviour represented in the simulation dataset.

Reflux Behaviour

The surrogate was also examined as the reflux ratio was varied from **3.00 to 4.35**.

The average predicted distillate purity increased from approximately **0.9360 at a reflux ratio of 3.00** to approximately **0.9517 at a reflux ratio of 4.05**. At a reflux ratio of 4.35, the average predicted value decreased slightly to approximately **0.9506**.

The average predicted bottoms purity increased overall from approximately **0.9489 at a reflux ratio of 3.00** to approximately **0.9610 at a reflux ratio of 4.35**.

The results therefore show an overall improvement in separation performance with increasing reflux, while also showing small non-monotonic variations in the predicted values. These variations are present within the finite DWSIM dataset and should not be interpreted as a perfectly monotonic relationship over every individual operating condition.

Overall, the physical validation showed that the surrogate predictions remained within reasonable ranges and reproduced the main behaviour observed in the DWSIM simulations.

12. Overfitting and Underfitting

The very high accuracy of Polynomial Regression requires some caution when interpreting the results.

Polynomial Regression achieved test-set R^2 values between **0.999963 and 1.000000**, while the 5-fold cross-validation results remained similarly high. The small difference between the test-set and cross-validation performance provides no strong indication of conventional train-test overfitting within the available dataset.

However, the dataset contains only **100 simulation cases**, and three input variables—feed benzene mole fraction, number of stages, and feed stage—were kept constant.

Consequently, the high accuracy demonstrates strong interpolation performance within the simulated operating region, but it does not establish that the model will generalize to substantially different column configurations.

Random Forest showed slightly lower accuracy but still performed well across the investigated operating region.

SVR showed substantially lower performance for some outputs, particularly distillate purity and the two heat-duty targets. This suggests that the selected SVR configuration was less effective at representing the relationships present in this dataset.

The model should therefore be regarded as a high-accuracy surrogate for the investigated simulation space rather than as a universal distillation model.

13. Final Model Selection

Based on the test-set accuracy, 5-fold cross-validation, and physical consistency checks, **Polynomial Regression was selected as the final surrogate model.**

It achieved the best predictive performance for every target variable.

Its test-set performance was:

- **xD:** $R^2 = 0.999963$
- **xB:** $R^2 = 0.999968$
- **QC:** $R^2 = 1.000000$

- **QR:** $R^2 = 1.000000$

The corresponding MAE and RMSE values were also substantially lower than those obtained using Random Forest and SVR.

The cross-validation results further support this selection. Polynomial Regression maintained mean R^2 values above **0.9997 for all four outputs**, with very small variation between folds.

Random Forest was the second-best model and demonstrated strong performance, particularly for bottoms purity and the heat-duty predictions. However, its errors were consistently higher than those of Polynomial Regression.

SVR produced the lowest overall performance and showed greater variation between folds, particularly for distillate purity.

Therefore, within the operating region represented by the DWSIM dataset, **Polynomial Regression provides the best combination of predictive accuracy, stability, and physical consistency.**

14. Limitations

The main limitation of the study is the size and coverage of the dataset.

The final dataset contains **100 DWSIM simulations**. The cases cover variation in feed temperature, feed pressure, reflux ratio, and bottoms withdrawal rate, while feed benzene mole fraction, number of stages, and feed stage were kept constant.

As a result, the surrogate model should not be considered a universal model for all possible benzene–toluene distillation configurations.

The model was developed for the following operating region:

- Feed temperature: **351.15–355.90 K**
- Feed pressure: **99,000–102,000 Pa**
- Feed benzene mole fraction: **0.50**
- Number of stages: **10**
- Feed stage: **5**

- Reflux ratio: **3.00–4.35**
- Bottoms withdrawal rate: **13.70–13.97 mol/s**

Predictions outside these ranges involve extrapolation and should therefore be validated against additional DWSIM simulations before being used.

A larger future dataset could improve the coverage of the operating space by varying feed composition, number of stages, and feed stage location while maintaining physically meaningful and converged simulations.

15. Conclusion

A machine-learning-based surrogate model was developed for a binary benzene–toluene distillation column using simulation data generated in DWSIM with the **Peng–Robinson thermodynamic model**.

A dataset of **100 successful DWSIM simulations** was generated and used to train three machine learning approaches: Polynomial Regression, Random Forest, and Support Vector Regression.

The models predicted four important column performance variables:

- Distillate purity (xD)
- Bottoms purity (xB)
- Condenser heat duty (QC)
- Reboiler heat duty (QR)

Polynomial Regression produced the highest predictive accuracy for all four targets. It achieved test-set R^2 values of **0.999963 for xD, 0.999968 for xB, and 1.000000 for both heat-duty targets**.

The 5-fold cross-validation results confirmed the stability of the model, with mean R^2 values above **0.9997 for all four outputs**.

Physical validation showed that **100 out of 100 predictions passed the validity checks for each target**. The predicted ranges were extremely close to the corresponding DWSIM ranges, and the reflux-ratio analysis reproduced the overall increase in separation performance observed within the investigated operating region.

Based on the combined statistical and physical evaluation, **Polynomial Regression was selected as the final surrogate model.**

The model provides a highly accurate approximation of the DWSIM results within the investigated operating space and can therefore be used to obtain rapid predictions without running a new rigorous simulation for every operating condition.

At the same time, the limited dataset size and fixed values of feed composition, number of stages, and feed stage restrict the model's generalization beyond the simulated region. The results should therefore be interpreted as a surrogate model for the investigated operating space rather than as a universal model of benzene–toluene distillation.

The final project package contains the DWSIM-generated dataset, source code, model evaluation results, cross-validation results, physical validation, and plots required to reproduce and assess the surrogate modelling workflow.